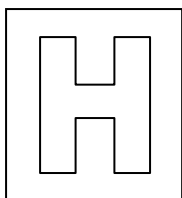


Candidate Name: \_\_\_\_\_

Class    Adm No

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## 2016 Preliminary Examination II

### Pre-university 3

**Biology Higher 2**

**9648/01**

**Paper 1 Multiple Choice**

**23 September 2016**

**1 hour 15 minutes**

Additional Materials: Optical answer sheet

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#### **READ THESE INSTRUCTIONS FIRST**

**Do not open this booklet until you are told to do so.**

Write your name, Adm No. and class on all the papers you hand in.

Do not use staples, paper clips, highlighters, glue or correction fluid.

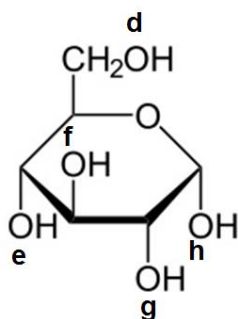
#### **Paper 1**

There are **forty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Multiple Choice Answer Sheet.

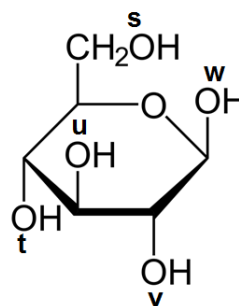
Calculators may be used.

Answer **all** questions

1. The diagram shows two isomers of a hexose labelled **I** and **II**. Four possible bonding positions are labelled **d**, **e**, **f**, **g** and **h** on one isomer, and **s**, **t**, **u**, **v** and **w** on another isomer.



**Isomer I**

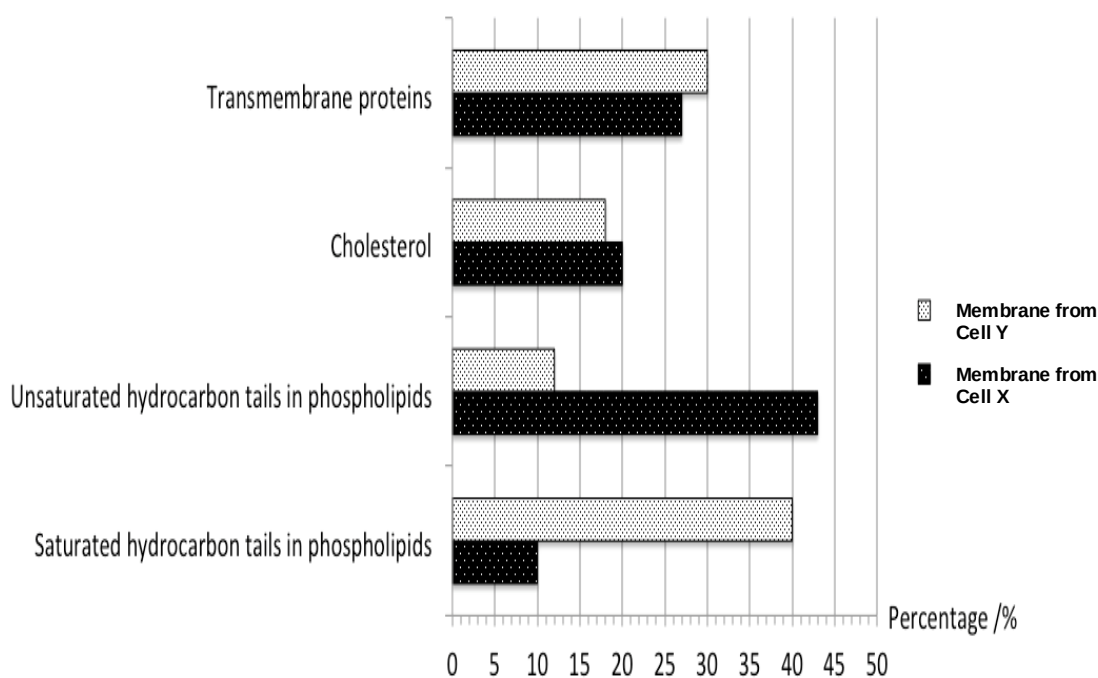


**Isomer II**

Which isomer and bonding positions are involved in the formation of amylopectin?

|          | Repeating units | Bonding position          |
|----------|-----------------|---------------------------|
| <b>A</b> | Isomer I        | <b>d-g</b> and <b>d-h</b> |
| <b>B</b> | Isomer I        | <b>e-h</b> and <b>d-h</b> |
| <b>C</b> | Isomer II       | <b>s-v</b> and <b>s-w</b> |
| <b>D</b> | Isomer II       | <b>t-w</b> and <b>s-w</b> |

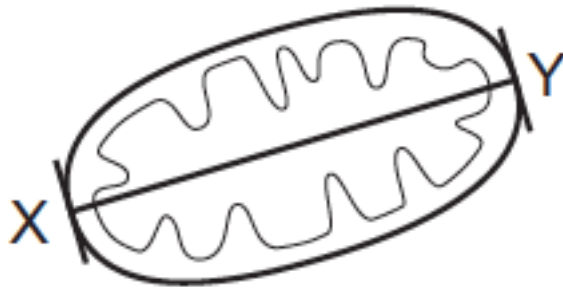
2. Reindeer are well adapted to survive extreme cold winters. One of these adaptations is the cell membrane composition at different parts of its body. The graph below shows the percentage composition of cell surface membrane components of Cell X and Cell Y taken from two different parts of the reindeer's body.



Which of the following statement best explains the differences in the membrane composition in Cell X and Cell Y?

- A Cholesterol decreases the membrane fluidity and prevents the membrane from breaking up by restraining the movement of phospholipids.
- B Cell X is taken from a lower part of the reindeer's leg as the unsaturated hydrocarbon tails in the cell surface membrane will prevent the fatty acids from packing close to each other.
- C Cell Y is taken from a lower part of the reindeer's leg as the saturated hydrocarbon tails in the cell surface membrane will prevent the fatty acids from packing close to each other.
- D Transmembrane proteins maintain the osmotic balance between the interior and exterior of the cell, hence preventing the cell membrane from solidifying at low temperatures.

3. What is the role of a biological catalyst in a metabolic reaction?
- A Increases both the activation energy and the energy yield.
  - B Decreases the activation energy and increases the energy yield.
  - C Decreases both the activation energy and the energy yield.
  - D Decreases the activation energy and has no influence on the energy yield.
4. The diagram shows a mitochondrion drawn from an electron micrograph.

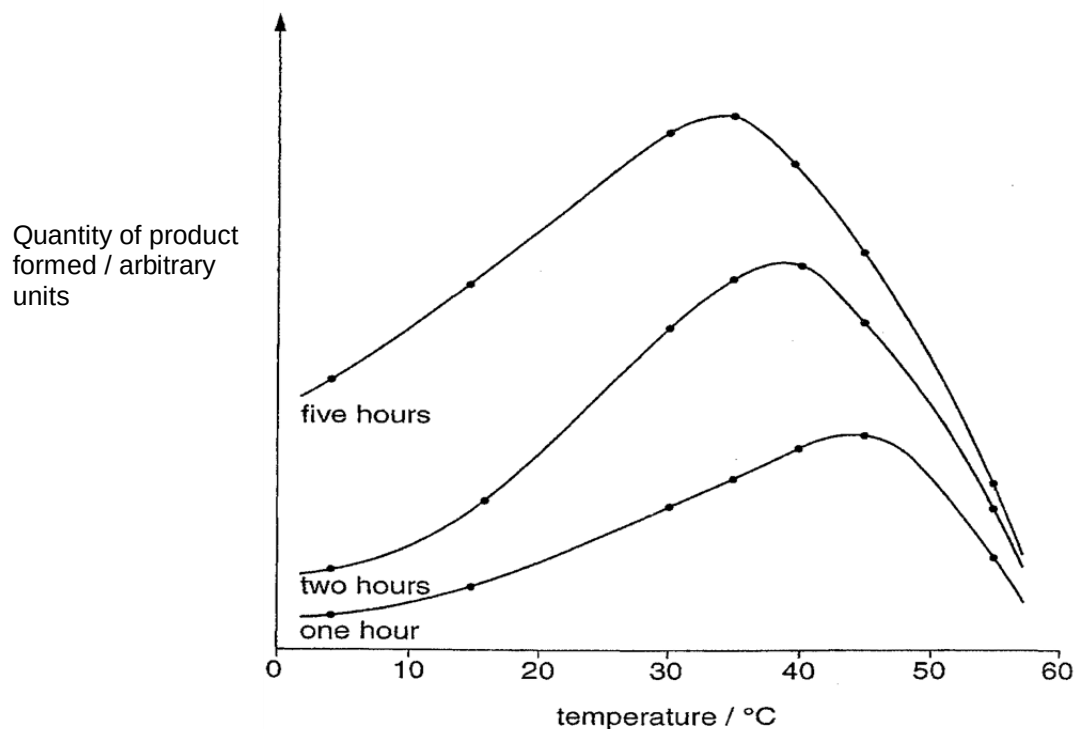


The length of the mitochondrion from X to Y is 3000 nm.

What is the magnification of the drawing of the mitochondrion?

- A X 200
- B X 2000
- C X 20000
- D X 200000

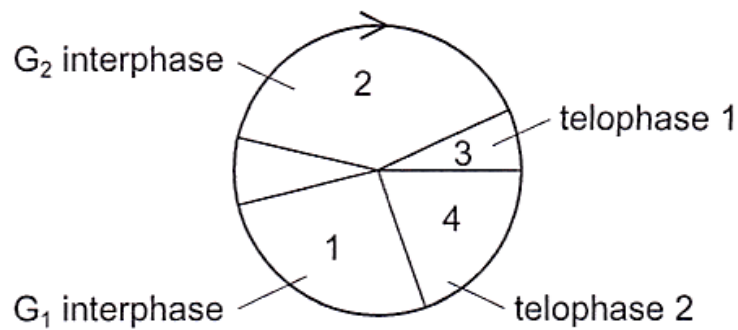
5. The following figure shows the results of an experiment, in which, samples containing the same concentration of enzyme and substrate were kept at different temperatures for periods of one, two and five hours. The quantities of product formed were then determined.



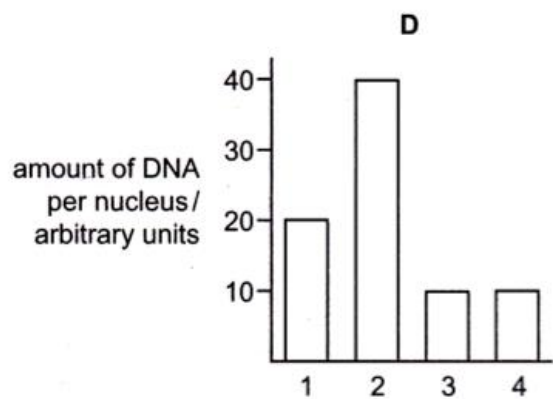
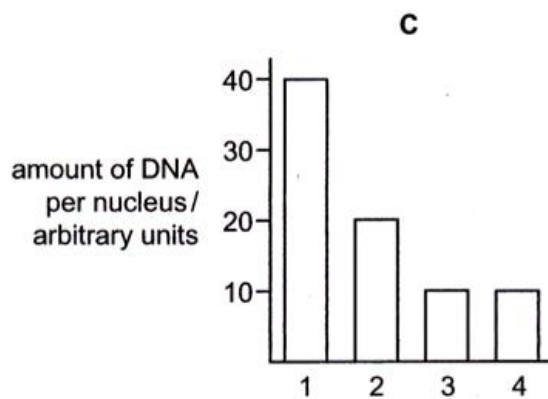
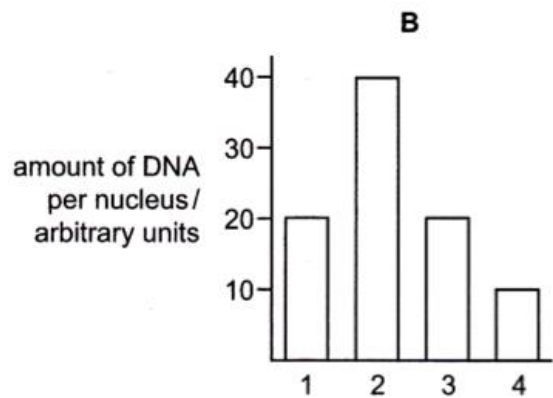
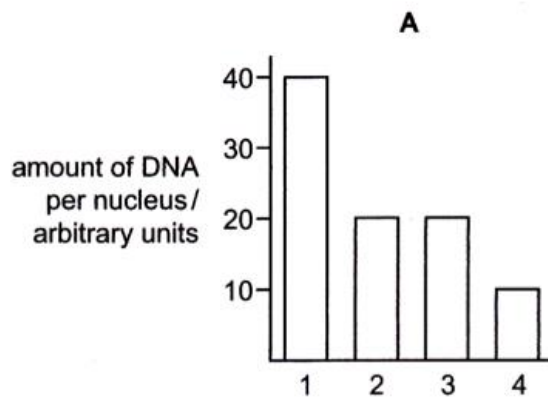
Which statement below does not describe the graphs?

- A As temperature increased, the quantity of products formed increased.
- B As the duration of the experiment decreased, the quantity of products decreased.
- C The enzyme is not active at 50°C for when the experiment is carried out for five hours.
- D Optimum temperature for the experiment held over one hour was highest as the enzyme had more disulfide bonds in stabilizing its structure.

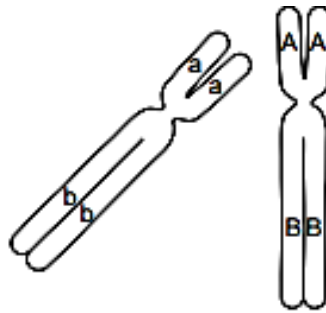
6. The amount of DNA per nucleus in a mouse cell during a meiotic cell cycle, as shown below, was measured.



Which bar correctly represents the variation in DNA content?



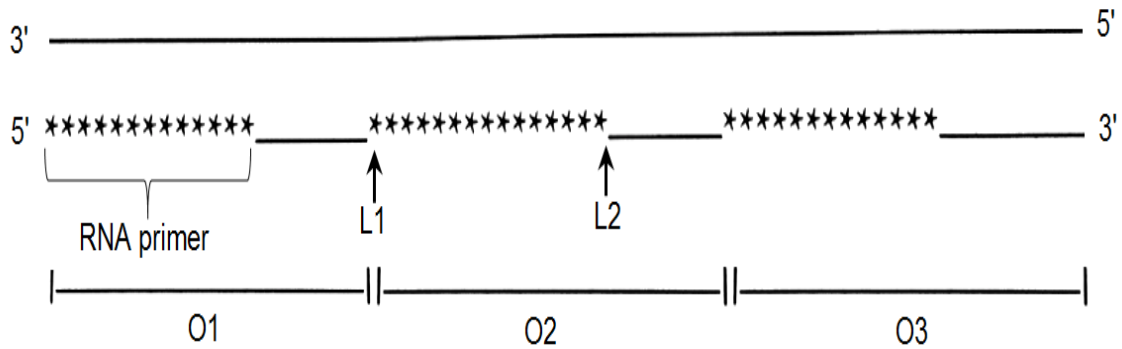
7. The diagram shows two homologous chromosomes in early prophase I of meiosis in a mouse cell. Two genes, A/a and B/b, whose loci occur on the homologous chromosomes are also shown.



Which option is a possible representation of these chromosomes as they progress from anaphase I to prophase II?

|   | anaphase I | prophase II |
|---|------------|-------------|
| A |            |             |
| B |            |             |
| C |            |             |
| D |            |             |

8. The diagram shows a DNA template with the lagging strand prior to the removal of the RNA primers.



Which row correctly shows the events taking place during the synthesis of the lagging strand?

|          | First Okazaki fragment synthesised | Site of phosphodiester bond formation catalysed by DNA ligase |
|----------|------------------------------------|---------------------------------------------------------------|
| <b>A</b> | O1                                 | L1                                                            |
| <b>B</b> | O1                                 | L2                                                            |
| <b>C</b> | O3                                 | L1                                                            |
| <b>D</b> | O3                                 | L2                                                            |

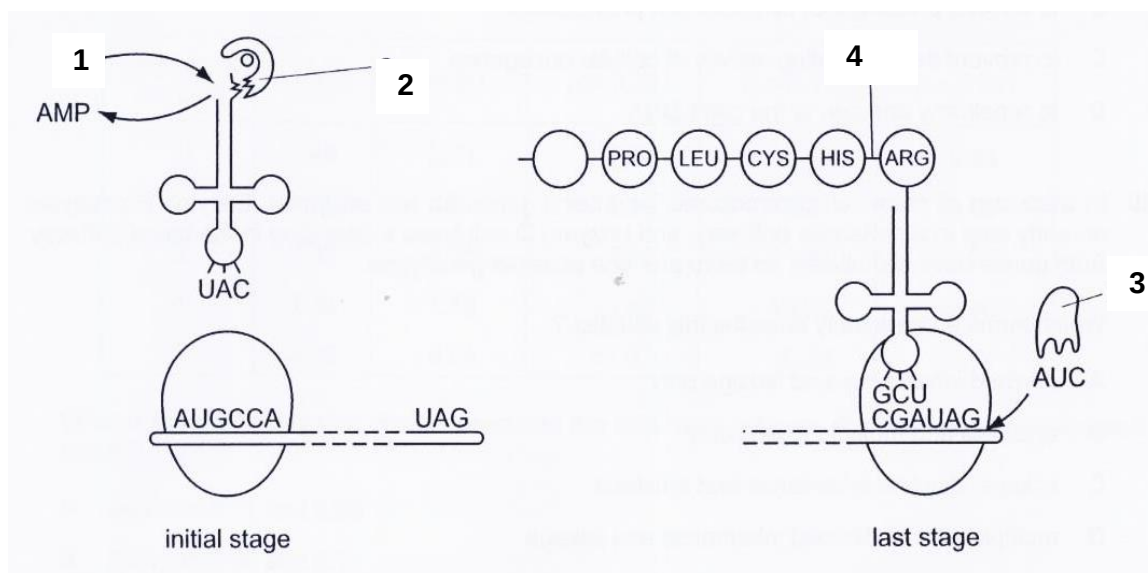
9. In a laboratory of Molecular Biology, the amino acids sequence of a cockroach intestine protein has been partially determined. The tRNA molecules used in the synthesis have the following anticodons:

3' UAC 5' 3' CGA 5' 3' GGA 5' 3' GCU 5' 3' UUU 5' 3' GGA 5'

What is the DNA nucleotide sequence of the non-template DNA strand?

- A** 5' ATG-GCT-GGT-CGA-AAA-CCT 3'  
**B** 5' ATG-GCT-CCT-CGA-AAA-CCT 3'  
**C** 5' ATG-GCT-GCT-CGA-AAA-GCT 3'  
**D** 5' ATG-GGT-CCT-CGA-AAA-CGT 3'

10. A number of molecules other than tRNA and mRNA are involved during translation.



Which line in the table is correct for labels 1 – 4?

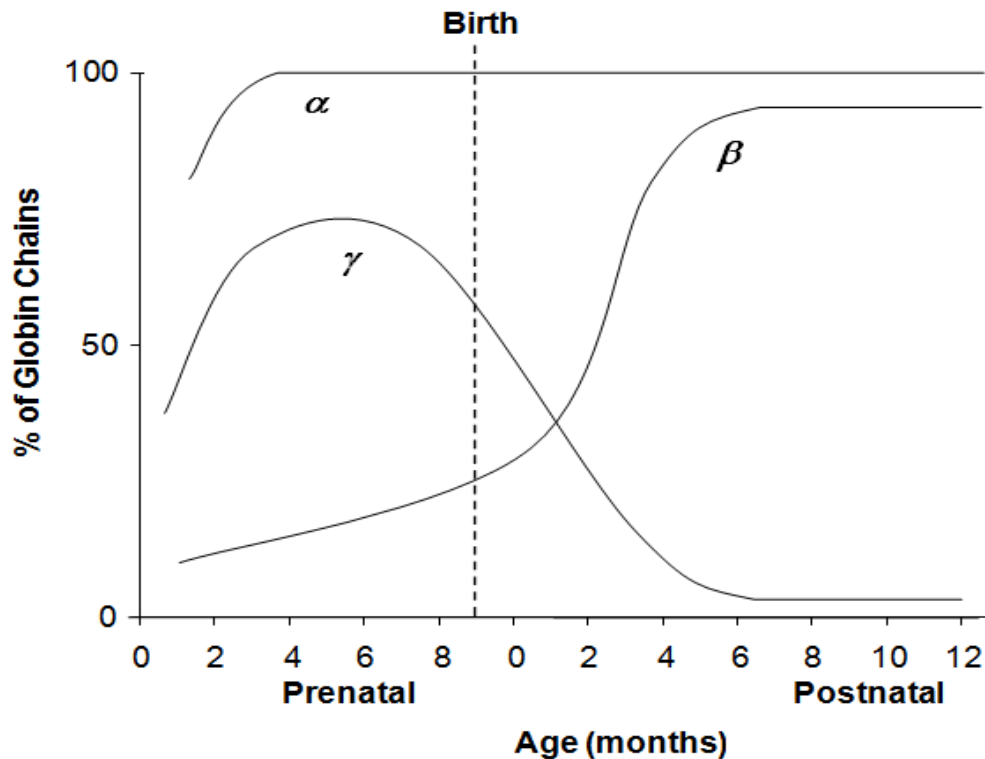
|   | 1   | 2                         | 3              | 4             |
|---|-----|---------------------------|----------------|---------------|
| A | ADP | Aminoacyl tRNA synthetase | Amino acid     | Hydrogen bond |
| B | ADP | Amino acid                | Release factor | Hydrogen bond |
| C | ATP | Aminoacyl tRNA synthetase | Release factor | Peptide bond  |
| D | ATP | Amino acid                | Aminoacyl tRNA | Peptide bond  |

11. Which of the following statements is true about prokaryotic plasmids?

- A Plasmids replicate along with the chromosomal DNA by using chromosomal DNA as a template.
- B Antibiotic-resistant genes on plasmids allow them to survive in antibiotic culture medium.
- C Plasmid DNA strands run in opposite directions and the copy number varies with the type of plasmid.
- D Binary fission ensures that plasmids are equally separated into daughter cells.

12. The globin gene family in humans consists of the  $\alpha$ ,  $\beta$  and  $\gamma$  genes. These genes code for the globin chains that make up haemoglobin and are expressed at different levels during different developmental stages.

The graph shows the expression of the various globin chains during the prenatal (fetal) and postnatal (after birth) periods.



Which statement cannot account for the differences in the levels of expression of globin chains?

- A Methyl groups are added to regulatory sequences of  $\gamma$ -globin gene during the postnatal period, allowing for chromatin remodeling complexes to be recruited.
- B Alternative splicing occurs in the mature mRNA of the  $\alpha$ -globin and  $\beta$ -globin genes, resulting in differences in the rate of expression of globin chains during the prenatal period.
- C A growth factor triggers the expression of a transcription factor that increases the rate of  $\beta$ -globin gene expression during the postnatal period.
- D The shortening of poly(A) tail in the mRNA of  $\gamma$ -globin genes reduces its stability, resulting in a decrease in the rate of expression of  $\gamma$ -globin chains during the postnatal period.

13. An analysis of the cells of a cancer patient revealed the presence of abnormal amounts of Ras proteins.

Which of the following could not explain for this observation?

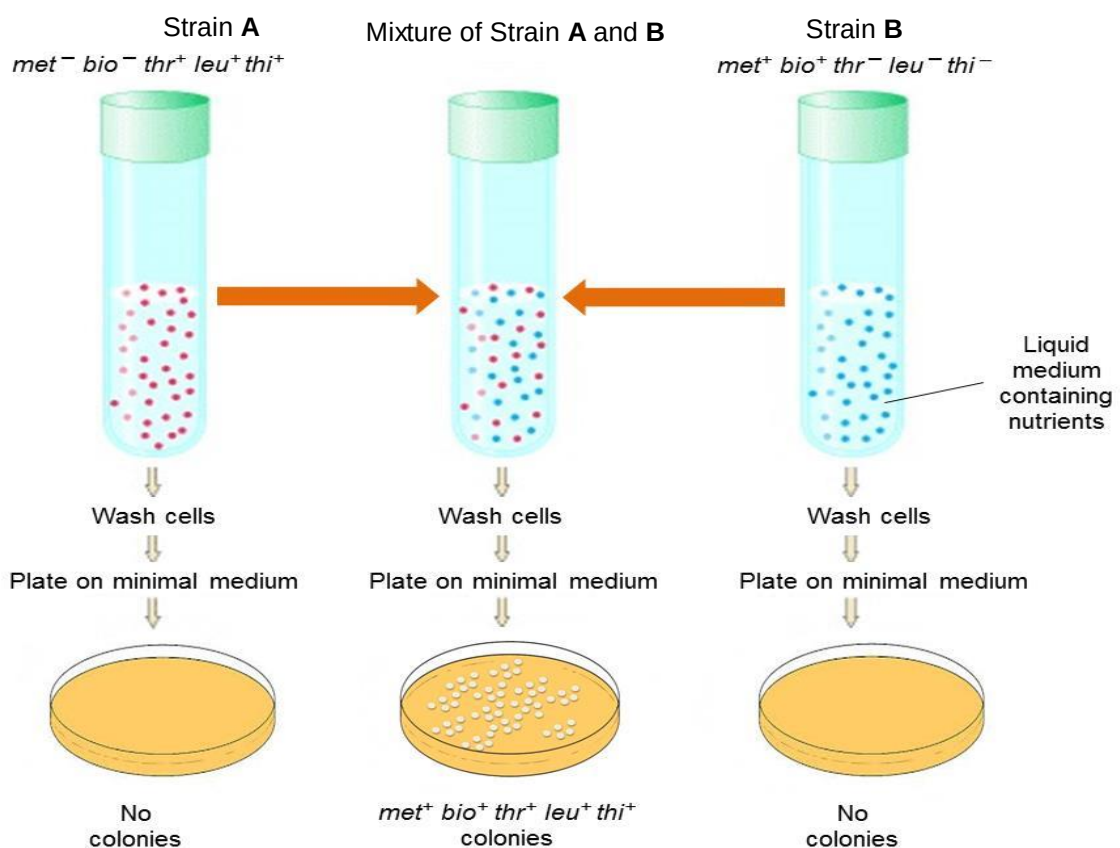
- A Gene amplification of Ras proto-oncogene
  - B A point mutation in a control element of the Ras proto-oncogene
  - C Translocation of Ras proto-oncogene to a region under the control of a more active promoter
  - D A single substitution in the exon of Ras proto-oncogene
14. A point mutation has occurred in *Escherichia coli*. Glucose and lactose are both absent in the culture medium where the mutant *E. coli* is grown in. An analysis of proteins synthesised by mutant *E. coli* found substantial amount of  $\beta$ -galactosidase, transacetylase and permease.

Which of the following mutations could have taken place in the *E. coli* cell?

- 1 Mutation in the operator of the lac operon
  - 2 Mutation in the *lacI* gene
  - 3 Mutation in the promoter of the lac structural genes
- A 1 only
  - B 2 only
  - C 1 and 2 only
  - D 2 and 3 only

15. In an experiment to study genetic recombination, two strains of *Escherichia coli* with different nutritional requirements were used. Strain **A** ( $met^- bio^- thr^+ leu^+ thi^+$ ) would grow on a minimal medium only if the medium were supplemented with methionine and biotin, while Strain **B** ( $met^+ bio^+ thr^- leu^- thi^-$ ) would grow on a minimal medium only if it was supplemented with threonine, leucine and thiamine.

Some of the dishes were plated only with Strain **A** bacteria, some only with Strain **B** bacteria, and some with a mixture of Strain **A** and Strain **B** bacteria that had been incubated together for several hours in a liquid medium.



Which conclusion can be best drawn from the experiment?

- A** The  $thr^+$ ,  $leu^+$  and  $thi^+$  genes in Strain **A** are located in its F plasmid.
- B** Cell-to-cell contact is necessary for the exchange of alleles between strain **A** and strain **B**.
- C** All the five genes are located on the F plasmid.
- D** Mutations resulted in  $thr^+$ ,  $leu^+$  and  $thi^+$  genes in Strain **B** to produce colonies on a minimal medium plate.

- 16.** The human immunodeficiency virus (HIV) has a highly specific gp120 glycoprotein that only allows it to infect cells with a CD4 receptor. To increase the host range of HIV in a laboratory setting, scientists often modify these viruses by artificially enclosing the viral nucleocapsid with a viral envelope with VSV-G glycoproteins embedded. VSV-G binds to LDL receptor, found ubiquitously on the cell membranes of many cell types, including kidney and liver cells that HIV do not usually infect.

A scientist has taken the following experimental steps:

1. Extracted an original and unmodified HIV nucleocapsid from wild-type HIV
2. Artificially enclose HIV nucleocapsid in a viral envelope containing VSV-G glycoproteins
3. Infect liver cells
4. Extract only newly synthesized virus particles released and examined them

What can the scientist expect to find in most newly synthesized virus particles extracted?

|          | <b>Viral envelope</b> | <b>Viral Genome</b> |
|----------|-----------------------|---------------------|
| <b>A</b> | VSV-G                 | VSV-G gene          |
| <b>B</b> | VSV-G                 | gp120 gene          |
| <b>C</b> | gp120                 | VSV-G gene          |
| <b>D</b> | gp120                 | gp120 gene          |

17. During summer, an isolated population of bighorn ram on a mountain has been captured and the length of their curled horns was measured over 30 years. Horn length of bighorn ram follows a distribution similar to that of its weight.

Which of the following statements describing the features of horn length in bighorn sheep are not correct?

- 1 Horn length is controlled by multiple alleles with different degrees of dominance.
- 2 Horn length shows quantitative expression with overlaps between categories.
- 3 Horn length is polygenic where inherited individual alleles have an additive effect.
- 4 Horn length is not affected by the environment

- A 1 and 2 only  
B 1 and 4 only  
C 2 and 3 only  
D 3 and 4 only

18. Which of the following causes variation in both sexually and asexually reproducing organisms?

- A Mutation  
B Polygenic inheritance  
C Crossing over  
D Independent assortment

19. Possession of white or coloured feathers in chickens is controlled by two genes **P/p** and **Q/q**. The phenotypes of offspring that are expected from mating two chickens, each of which is heterozygous at both loci, are shown in the Punnett square.

| Gametes   | <b>PQ</b>      | <b>Pq</b>      | <b>pQ</b>         | <b>pq</b>         |
|-----------|----------------|----------------|-------------------|-------------------|
| <b>PQ</b> | White feathers | White feathers | White feathers    | White feathers    |
| <b>Pq</b> | White feathers | White feathers | White feathers    | White feathers    |
| <b>pQ</b> | White feathers | White feathers | Coloured feathers | Coloured feathers |
| <b>pq</b> | White feathers | White feathers | Coloured feathers | White feathers    |

What best explains the proportion of white to coloured feathers in the Punnett square?

- A Dominant epistasis in which the epistatic allele is **P**
  - B Dominant epistasis in which the epistatic allele is **Q**
  - C Recessive epistasis in which the epistatic allele is **p**
  - D Recessive epistasis in which the epistatic allele is **q**
20. There are two hypotheses to explain the production of white, pale blue or dark blue flowers in a species of plant.

Hypothesis 1: There are two codominant alleles for flower color

Hypothesis 2: There are three alleles, one for each flower colour.

Which procedure is the best way of testing these hypotheses?

- A Analysis of the flower pigments in several different flowers by chromatography to find whether some plants contain more than one pigment.
- B Controlled cross pollination of all the different colour varieties available, in all possible combinations, and recording the colours shown by the offspring.
- C Surveying large wild populations and finding the ratios of the different colours in these.
- D Controlled self-pollination of several individuals of each of the colour varieties and recording the colours shown by the offspring of each individual plant sampled.

21. The following claims have been made about respiration.

- 1 Two turns of the citric acid cycle is required to oxidize 1 molecule of glucose.
- 2 Four molecules of carbon dioxide are generated for every molecule of acetyl CoA introduced into the Krebs Cycle.
- 3 During aerobic respiration, glucose produces pyruvate, CO<sub>2</sub> and ATP in the cytoplasm of a muscle cell.
- 4 Aerobic respiration can produce about 19 times the amount of ATP produced in anaerobic respiration.

Which of the following statements is/are true?

- A 1 only
- B 1 and 4 only
- C 2 and 4 only
- D 1, 3 and 4 only

22. *Rafflesia arnoldii* is a parasitic plant that produces the world's largest flower. It survives by invading the underground roots of vines to absorb nutrients essential for polypeptide synthesis. By doing so, it makes use of the nutrients to develop a giant and stinking flower that smells of rotting flesh in order to attract pollinating flies.

Which of the following two factors are most likely to act as limiting factors and inhibit growth of the vines when the *Rafflesia* parasite is present?

- 1 Carbon dioxide
- 2 Light
- 3 Photosynthetic enzymes
- 4 Water
- 5 Temperature

- A 1 and 2
- B 2 and 3
- C 3 and 4
- D 4 and 5

23. Which of the following gives an accurate comparison between intracellular receptors and cell surface receptors?

|          | Intracellular receptors                                                                      | Cell surface receptors                                                                   |
|----------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| <b>A</b> | May act as regulatory proteins and bind to DNA                                               | May catalyse the phosphorylation of intracellular proteins                               |
| <b>B</b> | Functions as the second messenger to activate other relay proteins                           | Binding of ligand always trigger the production of second messengers                     |
| <b>C</b> | Ligands can be water-soluble or lipid-soluble                                                | Ligands must be lipid-soluble                                                            |
| <b>D</b> | Made up of only hydrophobic amino acids to allow the interaction with lipid-soluble ligands. | Made up of hydrophobic amino acids which interact with the phospholipids of the membrane |

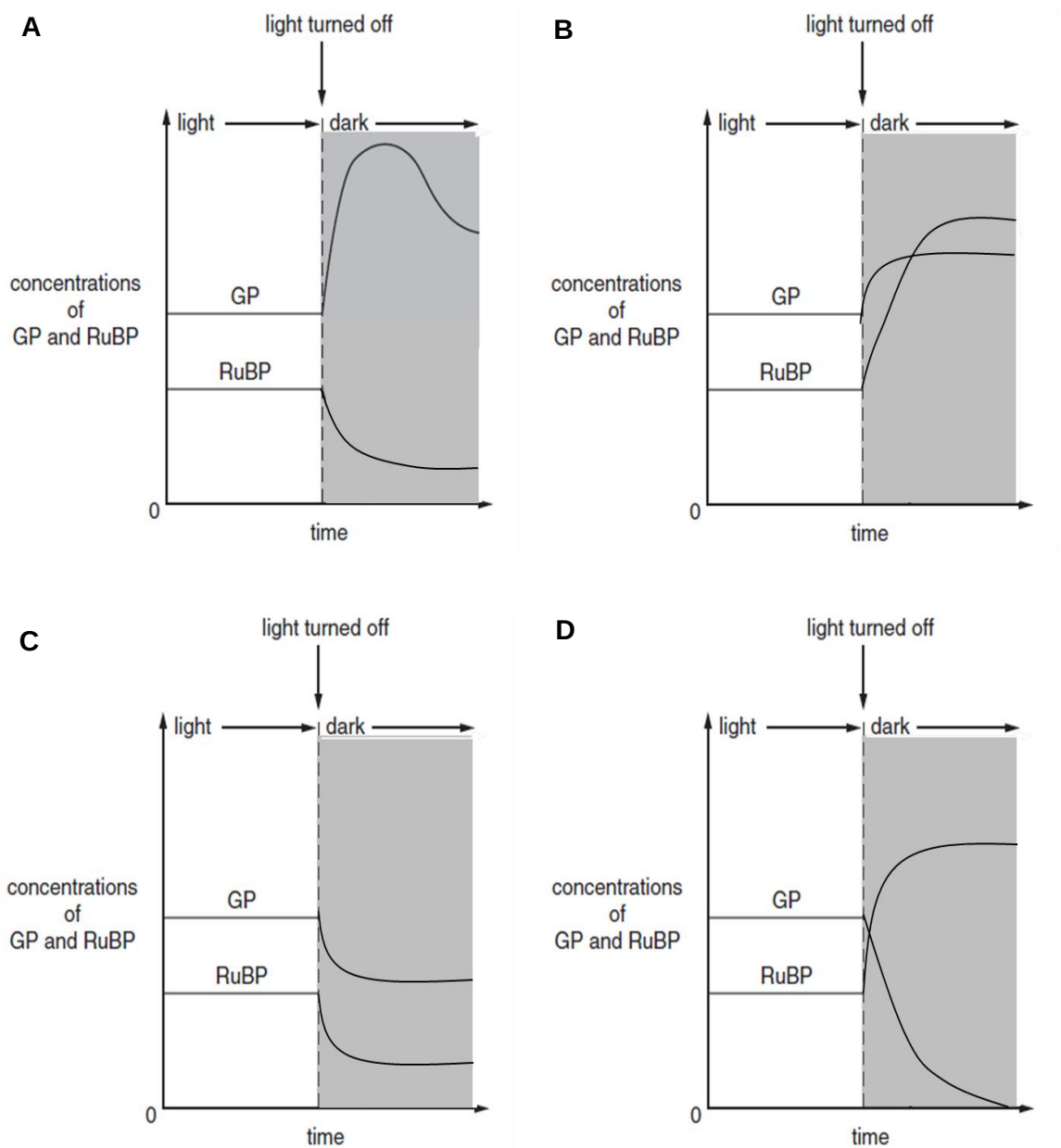
24. Which of the following statements about channel proteins is/are correct?

- I Channel proteins are either voltage-gated, ligand-gated or mechanically-gated.
- II Channel proteins do not undergo conformational changes when transporting molecules.
- III Channel proteins contain a hydrophilic pore lined with hydrophilic amino acids.
- IV Channel proteins serve to transport hydrophilic molecules of all sizes across the cell membrane.

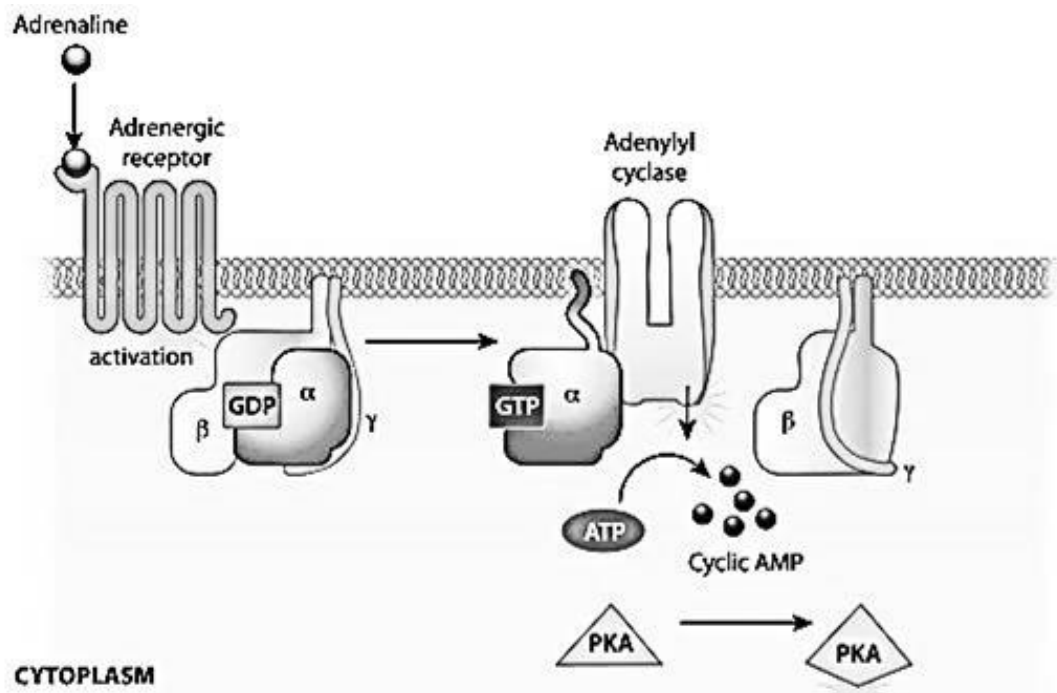
- A** I and II only
- B** II and III only
- C** I, II and III only
- D** All of the above

25. Concentrations of glycerate-3-phosphate (GP) and ribulose biphosphate (RuBP) were measured from samples of actively photosynthesising green algae in an experimental chamber.

Which of the following graphs show how the concentration of these compounds changes when the light source was turned off?



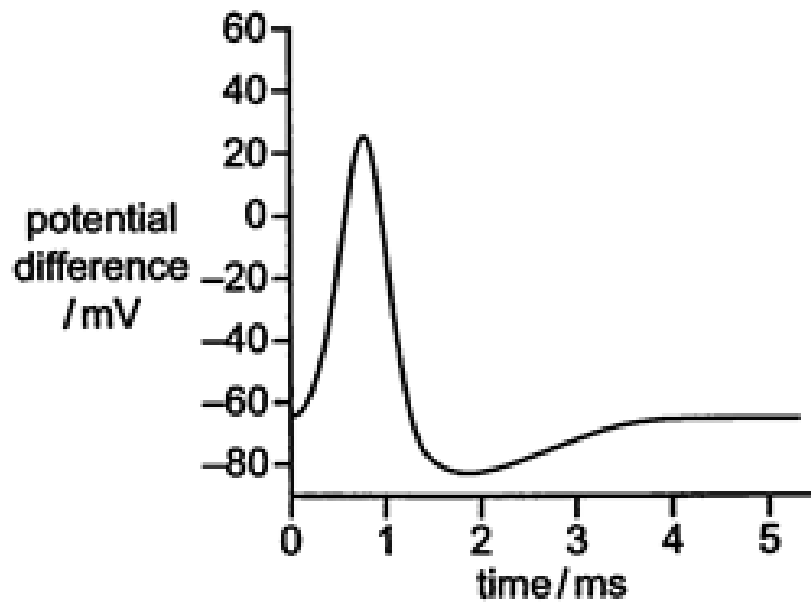
26. The following diagram shows the activation of the G protein-coupled receptor (GPCR) by the binding of adrenaline to the receptor. A mutation leads to constitutive signal transduction.



Which of the following is a possible result of the mutation?

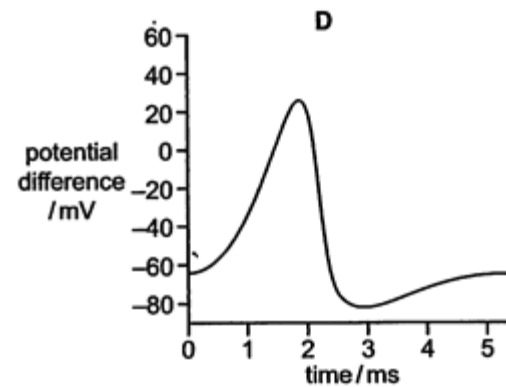
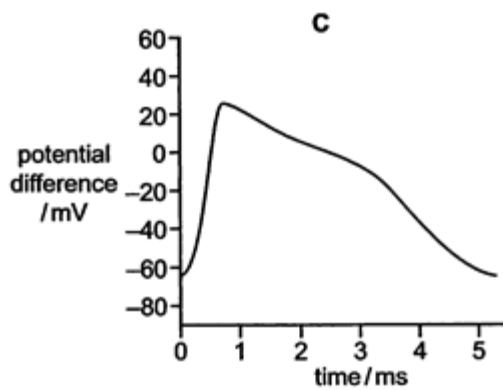
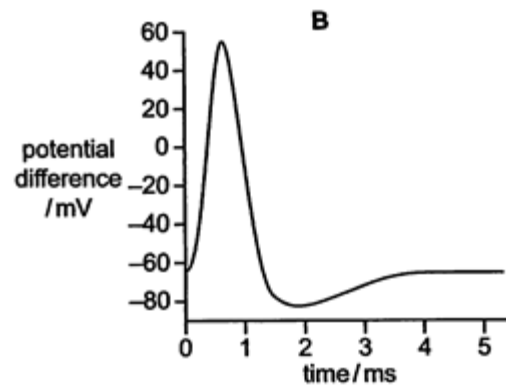
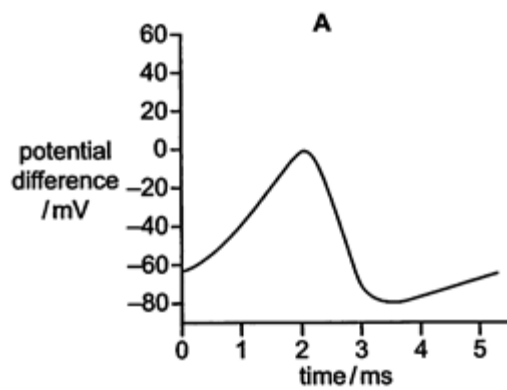
- A Conformational change in adenylyl cyclase such that it cannot convert ATP to cyclic AMP.
- B Adrenaline not being able to bind to the receptor.
- C Cyclic AMP not being able to bind to PKA.
- D GTPase in G protein failing to hydrolyse GTP to GDP.

27. The diagram below shows a normal action potential.



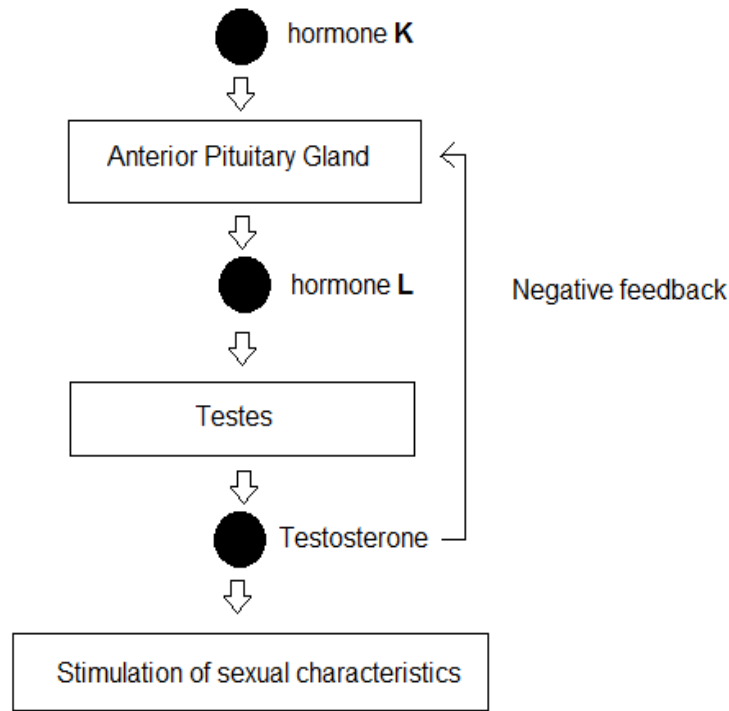
Drug X can bind and block voltage-gated  $K^+$  channels.

Which of the following diagrams shows an action potential in a neurone affected by such a drug?



28. Which is a correct description of the role of calcium ions in synaptic signalling?
- A Calcium ions are moved in by diffusion through ligand gated ion channels in pre-synaptic membranes of excitatory neurons, causing vesicles to move towards the pre-synaptic membrane as an impulse arrives.
  - B Calcium ions are moved in by diffusion through voltage gated ion channels in pre-synaptic membranes of excitatory neurons, causing vesicles to move towards the pre-synaptic membrane as an impulse arrives.
  - C Calcium ions are moved in by active transport through calcium pumps in pre-synaptic membranes of excitatory neurons, causing vesicles to move towards the pre-synaptic membrane as an impulse arrives.
  - D Calcium ions are moved out by active transport through calcium pumps in pre-synaptic membranes of excitatory neurons, causing vesicles to move towards the pre-synaptic membrane as an impulse arrives.

29. The production of testosterone is regulated by the anterior pituitary gland. Upon stimulation by hormone K, the gland releases hormone L which causes the testes to produce testosterone. The elevated level of testosterone exerts a negative feedback on the anterior pituitary gland.



Which of the following would be observed if a man consumes large amount of steroids containing testosterone?

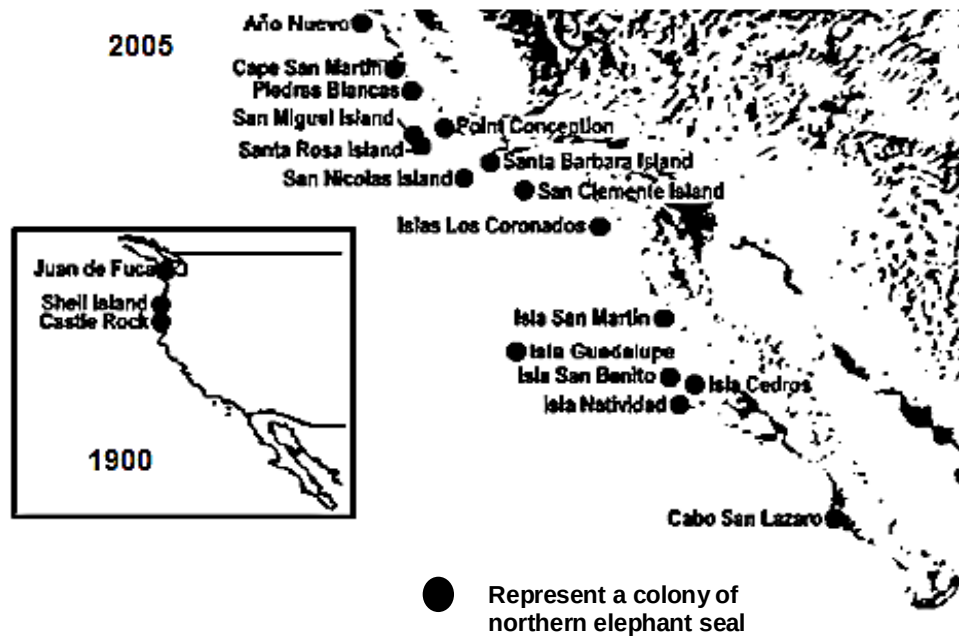
|          | Production of hormone K | Production of hormone L | Production of testosterone | Stimulation of sexual characteristics |
|----------|-------------------------|-------------------------|----------------------------|---------------------------------------|
| <b>A</b> | Decrease                | Decrease                | Decrease                   | No change                             |
| <b>B</b> | Decrease                | Decrease                | No change                  | Increase                              |
| <b>C</b> | No change               | Decrease                | Decrease                   | Increase                              |
| <b>D</b> | No change               | Decrease                | Decrease                   | No change                             |

30. Which of the following is not a limitation of the use of fossil records as evidence for evolution?
- A Fossils are damaged and incomplete.
  - B Some organisms may not form fossils.
  - C Fossils are found in different sedimentary rock layers.
  - D Fossils present in inaccessible areas are not available to us for study.
31. The following statements relate to molecular phylogenetics.
- 1 Lines of descent from a common ancestor to present-day organisms have undergone similar and fixed rates of DNA mutation.
  - 2 Organisms with similar base sequences in their DNA are closely related to each other.
  - 3 The number of differences in the base sequences of DNA of different organisms can be used to construct evolutionary trees.
  - 4 The proportional rate of fixation of mutations in one gene relative to the rate of fixation of mutations in other genes stays the same in any given line of descent.

Which statements, when taken together, suggest the existence of a 'molecular clock' that enables scientists to estimate the time at which one species might have diverged from another?

- A 1 and 2 only
- B 1 and 4 only
- C 2 and 3 only
- D 3 and 4 only

32. The northern elephant seal almost became extinct in the late 1800's following harvesting by whalers and sealers for their blubber which contains oil. A small colony of between 20 and 100 individuals survived on Guadalupe Island off Baja California. This colony has since given rise to about 160,000 elephant seals on the Pacific coast today. The following figure compares the distribution of elephant seals between 1900 and 2005.

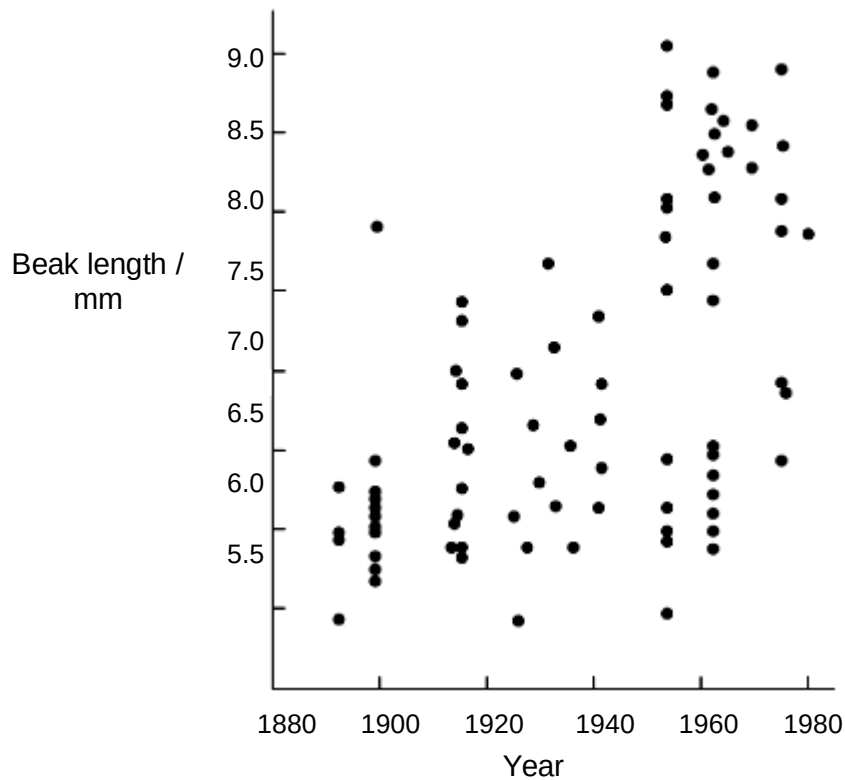


Which evolutionary mechanism could best explain why the degree of homozygosity is higher in the recent population of elephant seals compared to the original population in the 1800s?

- A Directional selection
- B Bottleneck effect
- C Founder effect
- D Genetic drift

33. The soapberry bug, *Jadera haematoloma*, uses its long beak to penetrate the fleshy fruit of the native soapberry tree to feed on the seeds at the centre. The bug also feeds on the fruit of the introduced golden rain tree.

Investigators measured the beak length of the soapberry bugs over eighty years. The results are shown in the graph.



Which of the following statements is a reasonable conclusion based on the above information?

- A The golden rain tree was introduced around 1970.
- B The change in beak length is an example of stabilizing selection.
- C The diameter of the golden rain tree fruit acted as a selection pressure on beak length.
- D The response of an individual golden rain tree to predation by soapberry bugs would be to grow larger fruit.

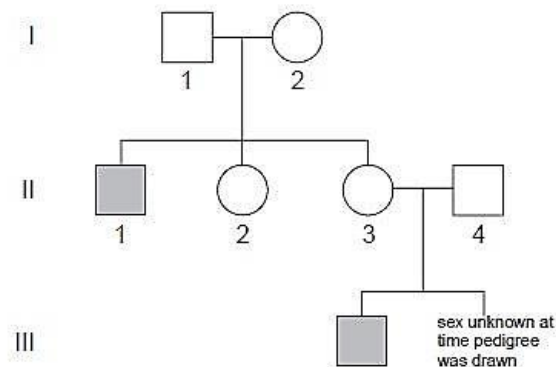
- 34.** EcoRI is a restriction enzyme produced by *Escherichia coli*. It recognises the sequence GAATTC and is able to make a staggered cut in the DNA. *E. coli* is susceptible to attack by bacteriophages.

Which of the following correctly describes the natural function of EcoRI?

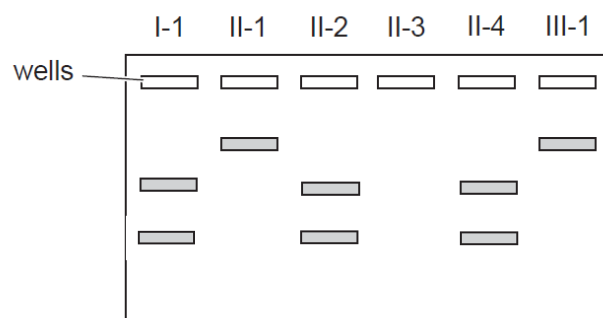
- A** It lowers infectivity rate of *E. coli* by bacteriophages that have double-stranded DNA as their genetic material, since EcoRI can recognise and cleave specific viral nucleotide sequences.
- B** It cleaves *E. coli* DNA sequences to produce restriction fragments that can be packaged into bacteriophages for infection of new bacterial cells.
- C** It increases infectivity rate of *E. coli* by bacteriophages because it promotes cleavage of viral DNA at specific nucleotide sequences in order to package more viral DNA into new virus particles.
- D** It cleaves *E. coli* DNA sequences so that the bacterial cell which has been infected will undergo autolysis and self-destruct, preventing further infection of new bacterial cells.

35. Menkes' syndrome in humans is characterised by sparse and wiry hair, growth failure and deterioration of the nervous system. Onset of the Menkes' syndrome usually occurs during infancy.

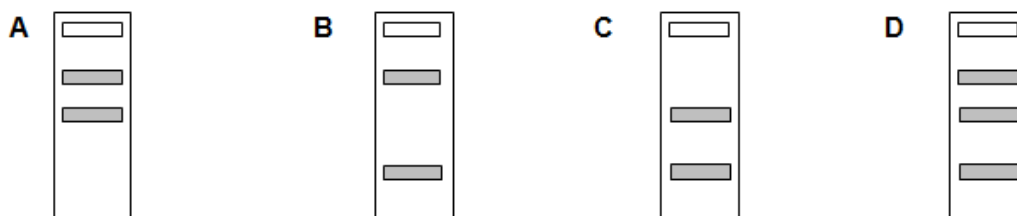
A family, in which this X-linked disorder was present, underwent Restriction Fragment Length Polymorphism (RFLP) analysis using gel electrophoresis. The family pedigree is shown below.



The RFLP analysis resulted in the following distribution of bands in the gel.



What would be the band pattern of individual II-3?



36. Which of the following options correctly describe the events occurring in each process?

|          | <b>Action of restriction enzyme</b> | <b>Ligation</b>                   | <b>Annealing</b>                  | <b>Denaturation</b>               |
|----------|-------------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| <b>A</b> | Breaking of hydrogen bonds          | Formation of hydrogen bonds       | Formation of phosphodiester bonds | Breaking of phosphodiester bonds  |
| <b>B</b> | Breaking of hydrogen bonds          | Formation of hydrogen bonds       | Formation of phosphodiester bonds | Formation of phosphodiester bonds |
| <b>C</b> | Breaking of phosphodiester bonds    | Formation of phosphodiester bonds | Formation of hydrogen bonds       | Breaking of hydrogen bonds        |
| <b>D</b> | Breaking of phosphodiester bonds    | Formation of phosphodiester bonds | Breaking of hydrogen bonds        | Formation of hydrogen bonds       |

37. Which of the following are ethical concerns arising from the Human Genome Project?

- 1 Scientists tracing migration of different population groups based on maternal inheritance
- 2 Genetic counsellors giving advice to people who are genetically pre-disposed to risks
- 3 Parents choosing to abort foetuses with minor disorders based on genetic testing results
- 4 Scientists developing tests for only some disease-causing genes
- 5 Employers refraining from hiring people with greater risk of developing genetic diseases

- A** 1 and 4  
**B** 2 and 5  
**C** 3 and 4  
**D** 3 and 5

38. Which of the statements about stem cells are false?

- 1 Embryonic stem cells are useful as they have the ability to differentiate into any cell type.
- 2 Embryonic stem cells are less easily isolated compared to neural crest stem cells.
- 3 Umbilical cord blood stem cells have the same developmental potential as neural crest stem cells.
- 4 Induced pluripotent stem cells undergo differentiation to give rise to various cell lineages.
- 5 Stem cells have the ability to self-renew without any stimulus.

- A 1, 2 and 5 only  
B 1, 4 and 5 only  
C 2, 3 and 4 only  
D 3, 4 and 5 only

39. One way to treat  $\beta$ -thalassaemia is to transplant bone marrow cells from a genetically compatible donor into a patient. A potential gene therapy involves adding the normal and dominant allele for  $\beta$ -globin to the patient's cells.

What would ensure that the normal gene is passed on to the next generation?

- A Using a retrovirus to introduce the normal  $\beta$ -globin gene into bone marrow cells.  
B Using an adenoviral vector to introduce the normal  $\beta$ -globin gene into bone marrow cells.  
C Using an adenoviral vector to introduce the normal  $\beta$ -globin gene into an egg cell.  
D Using a retrovirus to introduce the normal  $\beta$ -globin gene into an egg cell.

40. Which of the following is an example of genetically modified organisms?
- A Cows that grow to adult sizes quickly due to injection of recombinant bovine somatotropin
  - B Durians that ripen slower due to anti-sense technology
  - C High yielding rice that are flood resistant, due to intensive self-pollination over generations
  - D Cows that grow to abnormal size due to inbreeding

**End Of Paper**